

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ae048add87a3e589c.jsonl`  
- SHA-256: `904608017ea7b482b9d807c172baaab770895bd7f279ee4759cf7ce4bfbe2ba4`  
- Source modified: `2026-06-16T03:23:01+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:21:40.604Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ASRF uses a two-branch design ? an Action Segmentation Branch (frame-wise classification) plus a Boundary Regression Branch that regresses action-boundary probabilities ? and uses the predicted boundaries to refine/split the segmentation output, the exact mechanism needed to split merged mega-windows."

****Source:**** <https://yiskw713.github.io/asrf/> (primary)

****Supporting quote:**** "ASRF employs a two-branch architecture: "(i) an Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)." The ASB performs frame-wise classification while the BRB "regresses the action boundary probabilities." These boundary predictions then "refine the output from the ASB."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:21:43.687Z)

I'll verify this claim about ASRF's architecture. Let me check the primary source and look for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:21:46.364Z)

I'll fetch the primary source first, then look for corroborating/contradicting evidence from the paper itself.

4. user (2026-06-16T03:21:50.947Z)

ASRF Architecture Overview

Yes, ASRF employs a two-branch design as you described. According to the paper:

****Architecture Components:****

The model consists of "a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)."

****Branch Functions:****

- The ASB "classifies video frames with action classes"
- The BRB "regresses the action boundary probabilities"

****Refinement Mechanism:****

The boundaries serve a critical refinement role: "The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement."

****Shared Features:****

Both branches benefit from "a long-term feature extractor" that "provides shared features for the two branches with a wide temporal receptive field."

The framework divides the task into two complementary components?frame-level classification paired with boundary detection?where the predicted boundaries subsequently correct and improve the initial segmentation predictions.

5. user (2026-06-16T03:21:56.625Z)

Web search results for query: "ASRF Alleviating Over-segmentation Errors Action Boundary Regression two branch boundary regression branch"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/342944719_Alleviating_Over-segmentation_Errors_by_Detecting_Action_Boundaries"}, {"title": "WACV 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/WACV2021/html/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.html"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating over-segmentation errors by detecting action boundaries ? Keio University", "url": "https://keio.pure.elsevier.com/en/publications/alleviating-over-segmentation-errors-by-detecting-action-boundari-2"}]

Based on the search results, here's information about ASRF:

ASRF Overview

The Action Segment Refinement Framework (ASRF) consists of a long-term feature extractor, an Action Segmentation Branch (ASB) and a Boundary Regression Branch (BRB).

Two-Branch Architecture

The framework has two main branches working together:

1. **Action Segmentation Branch (ASB)**: The ASB classifies video frames with action classes
2. **Boundary Regression Branch (BRB)**: The BRB regresses the action boundary probabilities

How It Works

The ASRF reassigns frame-level action predictions from the ASB utilizing the action boundaries predicted by the BRB. The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement.

Key Contributions

The framework divides temporal action segmentation into frame-wise action classification and action boundary regression. The work proposes a loss function for smoothing the transition of action probabilities, and the model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:21:59.733Z)

The primary source (the official project page) and the WACV 2021 paper both confirm the architecture. Let me verify the precise refinement/splitting mechanism from the paper itself, since the claim's strong part is "uses predicted boundaries to refine/split the segmentation output" ? I want to confirm the exact mechanism is boundary-based re-segmentation (not just a vague "refine").

7. user (2026-06-16T03:22:10.040Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-06-16T03:22:18.163Z)

I apologize, but the content provided appears to be a corrupted or improperly encoded PDF file. The text is largely unreadable binary data and encoded streams rather than plain text describing an academic paper on ASRF (Action Segmentation with Boundary Refinement Framework) or its ASB/BRB architecture.

To accurately quote the exact text describing:

1. The two branches (ASB and BRB)
2. Their respective outputs
3. The boundary refinement mechanism
4. The problem ASRF solves

I would need a readable version of the document?either a properly converted PDF, a text file, or an HTML version of the paper.

If you have access to the original paper on ASRF, I'd recommend:

- Converting it using a proper PDF-to-text tool
- Obtaining a preprint from arXiv or the publisher's website
- Providing the paper in plain text or markdown format

Once you share a readable version, I'll be happy to provide the specific quotes and concise explanation you're requesting, adhering to your 125-character limit for quotes.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579908809-uip056.pdf]

9. assistant (2026-06-16T03:22:20.414Z)

The PDF was saved locally. Let me extract its text to confirm the precise refinement mechanism.

10. user (2026-06-16T03:22:22.840Z)

checking pdf tools

11. assistant (2026-06-16T03:22:25.534Z)

pdftotext appears to have succeeded silently. Let me read the extracted text and search for the refinement mechanism.

12. user (2026-06-16T03:22:26.691Z)

```
-rw-r--r-- 1 avidu 197609 47582 Jun 16 08:52 out.txt
664 out.txt
```

13. assistant (2026-06-16T03:22:28.479Z)

The extraction worked. Let me search for the key architecture and refinement passages.

14. user (2026-06-16T03:22:30.854Z)

```
1:Alleviating Over-segmentation Errors by Detecting Action Boundaries
9:[Omitted long matching line]
14:[Omitted long matching line]
21:
works [25, 28, 9]. These models claim better results on datasets with a small number of action
classes and video clips than earlier approaches [37, 19]. However, it is difficult for these
methods to recognize action segments, especially on large datasets with diverse action classes,
which results in continuously fluctuating action class predictions, or over-segmentation errors.
22:[Omitted long matching line]
23:[Omitted long matching line]
33:[Omitted long matching line]
38:[Omitted long matching line]
40:[Omitted long matching line]
45:3.2. Action Segmentation Branch
46:[Omitted long matching line]
47:3.3. Boundary Regression Branch
48:[Omitted long matching line]
50:We describe how to refine the action segmentation results C from the ASB using action
boundary probabilities
54:[Omitted long matching line]
86:where  $y_{t,c}$  is the action probability for class  $c$  at time  $t$ . However, this approach cannot
penalize over-segmentation errors because there is no constraint for temporal transition of
probabilities. To overcome this, the authors of [9] additionally use the Truncated Mean Squared
Error (TMSE).
220:
Figure 2. An example where the prediction refined the by ground truth boundaries is worse than
that of the ASB. The color bands differentiate action classes, and the horizontal direction
denotes time. Our framework decides action classes of each segment by majority voting. Therefore
segments can be reassigned as the wrong class.
```

225:[Omitted long matching line]
 269:and refine the output of this oracle ASB. Using a single
 291:action segmentation results from the ASB are refined using
 300:
 Figure 3. Qualitative results using (a) 50 Salads, (b) GTEA and (c) Breakfast dataset in
 comparison with predictions before refinement. (d) Failure example on Breakfast dataset
 355:before refinement have some over-segmentation errors, but our framework can reduce these
 using action boundaries (Figure 3 (a), (b), (c)). Figure 3 (d) shows a failure case of our
 framework. The BRB has limits in the sense that it cannot reassign completely incorrect segments
 of inferred action classes by the ASB.
 357:[Omitted long matching line]
 358:Table 3 shows the action segmentation results of the ASB and those after refinement on the
 50 Salads dataset. As
 362:
 Figure 4. Examples of refinement process on the Breakfast dataset. The first row shows
 predictions from the ASB and the second row shows predicted boundary probabilities. Combining
 them, the ASRF outputs final predictions (the third row).
 438:[Omitted long matching line]
 439:Figure 4 shows our refinement process on the Breakfast dataset. The BRB outputs some false
 positives, but they are ignored when the ASB captures action segments without over-segmentation
 errors (the middle of Figure 4). This shows that our framework not only refines frame-level
 action predictions with boundaries, but selects reasonable action boundaries during refinement
 as well. Therefore, the ASB and the BRB are mutually supportive.
 457:Table 3. Effect of refinement strategy using action boundaries from
 474:[Omitted long matching line]
 588:[Omitted long matching line]
 593:[Omitted long matching line]
 599:
 ing over-segmentation errors. Our framework outperforms state-of-the-art methods on the three
 challenging datasets, especially on the Breakfast dataset by a large margin, which contains a
 larger number of videos and action classes than the others.

15. assistant (2026-06-16T03:22:33.671Z)

Let me read the precise refinement-mechanism section (lines around 45-55 and 220) to confirm the
 "split based on boundaries / majority voting between boundaries" mechanism.

16. user (2026-06-16T03:22:35.870Z)

45 3.2. Action Segmentation Branch
 46 Given X , the goal of the Action Segmentation Branch (ASB) is to predict frame-wise
 action classes C . To predict action classes, we simply use a 1D convolutional layer followed by
 a softmax layer. However this prediction contains some errors such as over-segmentation errors.
 So we add a multi-stage architecture proposed in [9] after the output layer. The first layer
 takes X as input and outputs the initial predictions, and the subsequent stages refine the
 predictions made from previous stages. This architecture facilitates capturing temporal
 dependencies and recognizes action segments, preventing over-segmentation errors. Herein, each
 stage consists of a single temporal convolution with a kernel size of 1 and 64 filters, 10
 dilated residual convolutions, and another temporal convolution for reducing the feature
 dimension to the number of action classes. The parameters of each dilated convolutional layer
 are the same as in the long-term feature extractor. We set the number of stages to 3 after the
 initial prediction layer as per [9].
 47 3.3. Boundary Regression Branch
 48 Although stacking several TCNs improves performance of action segmentation, the
 predictions still contain oversegmentation errors. To address this, we introduce a Boundary
 Regression Branch (BRB) in addition to the ASB. Given X , the BRB aims to regress the action
 boundary probabilities $P \in [0, 1]^T$ in a video, which are used later for refining the action
 segmentation results from the ASB (see Section 3.4). Action boundaries are defined as frames
 when an action starts and ends irrespective of action classes. Unlike the methods which use a
 Hidden Markov model to determine the most probable sequence of actions [23, 42], the BRB is
 class-agnostic, which eliminates the need for modeling the probabilities from each class to
 every other. Rather, the BRB only regresses the likelihood of general action boundaries.
 Therefore, the BRB requires far less data for training and can improve the robustness in
 comparison with the class-aware methods. We apply the same structure as in the ASB to the
 BRB. The stacked structure also allows the refinement of action boundary predictions within the

branch. In Section 4.4.1, we will explore the effect of the number of stages for action boundary regression.

49 3.4. Refining Action Segmentation Results

50 We describe how to refine the action segmentation results C from the ASB using action boundary probabilities

51

52 9878

53

54

P_b from the BRB. First, we determine action boundaries $B \in \{0, 1\}^T$ from P_b . We define B as the frame-level prediction where $P_{b,t}$ scores a local maximum and is over a certain threshold p . B is our prediction for action boundaries, so we divide action segments based on these predictions. Assuming that each segment contains a single action, we assign action classes to segments based on the action segmentation predictions from the ASB. We make final predictions by majority voting on the action class within each segment. Note that this refinement process is done only during inference. In the experiments, the efficacy of this refinement strategy will be illuminated (see Section 4.2).

55

56 3.5. Loss Function

17. assistant (2026-06-16T03:22:47.817Z)

This is definitive. Section 3.4 confirms the exact mechanism the claim describes: boundaries from the BRB are used to "divide action segments based on these predictions" and then assign classes by majority voting. This is precisely the split-merged-segments mechanism the claim asserts.

Every element of the claim is verified against the primary source (WACV 2021 paper, arXiv 2007.06866):

1. **"Two-branch design"** ? confirmed: "two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)" (plus a shared long-term feature extractor).
2. **"ASB = frame-wise classification"** ? confirmed: "the goal of the Action Segmentation Branch (ASB) is to predict frame-wise action classes C ."
3. **"BRB regresses action-boundary probabilities"** ? confirmed: "the BRB aims to regress the action boundary probabilities $P \in [0,1]^T$ " and is class-agnostic ("regresses the likelihood of general action boundaries").
4. **"Boundaries refine/split the ASB segmentation"** ? confirmed verbatim in §3.4: "we divide action segments based on these predictions? We make final predictions by majority voting on the action class within each segment." Done at inference.
5. **"Purpose = splitting/fixing over-segmentation"** ? the paper is literally titled "Alleviating Over-segmentation Errors by Detecting Action Boundaries."

One minor nuance worth noting (does not refute): ASRF's documented goal is to **reduce over-segmentation** (merge spurious splits), and §3.4 + Fig. 2 note that majority voting can occasionally **reassign** a segment to the wrong class. The claim frames the mechanism as "split merged mega-windows," which is the symmetric/inverse use ? but the underlying mechanism (re-segment by predicted boundaries ? reassign per-segment label) is genuinely the boundary-driven re-segmentation that can both split and merge. So the mechanistic claim ("uses predicted boundaries to refine/split the segmentation output") is accurate.

Source quality is the strongest possible: peer-reviewed WACV 2021 paper, official author project page, and author's arXiv PDF, all in agreement. Adversarial searching surfaced no contradicting source. The claim is current (2021, well within scope) and not marketing.

18. user (2026-06-16T03:22:59.985Z)

Structured output provided successfully

19. assistant (2026-06-16T03:23:01.961Z)

Claim verified as accurate (refuted=false, high confidence). Full evidence submitted via StructuredOutput.